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| <b>Exp No: 5</b> | <b>Experiments on Packet capture tool: Wireshark</b> |
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### **AIM:**

To understand the features of Wireshark as a packet capture tool and to understand the encapsulation of information at the various layers of a protocol stack.

### **ALGORITHM / PROCEDURE:**

1. Install and open Wireshark. The home screen lists all the network interfaces with a small activity graph.
2. Double click the active interface (Ethernet / Wi-Fi) to start the capture.
3. Generate some traffic: open a website in the browser, run ping google.com and nslookup example.com in the Command Prompt.
4. Stop the capture using the red square button.
5. Apply display filters in the filter bar, for example icmp, dns, tcp.port == 80, ip.addr == 192.168.1.25.
6. Select one packet and expand its layers in the **Packet Details** pane. Study the Frame, Ethernet II, Internet Protocol, TCP / ICMP headers.
7. Select any header field and see the corresponding bytes highlighted in the **Packet Bytes** pane.
8. Use **Analyze > Follow > TCP Stream** to follow the full TCP conversation and note the three-way handshake (SYN, SYN-ACK, ACK).
9. Save the capture using **File > Save As (.pcapng)**.

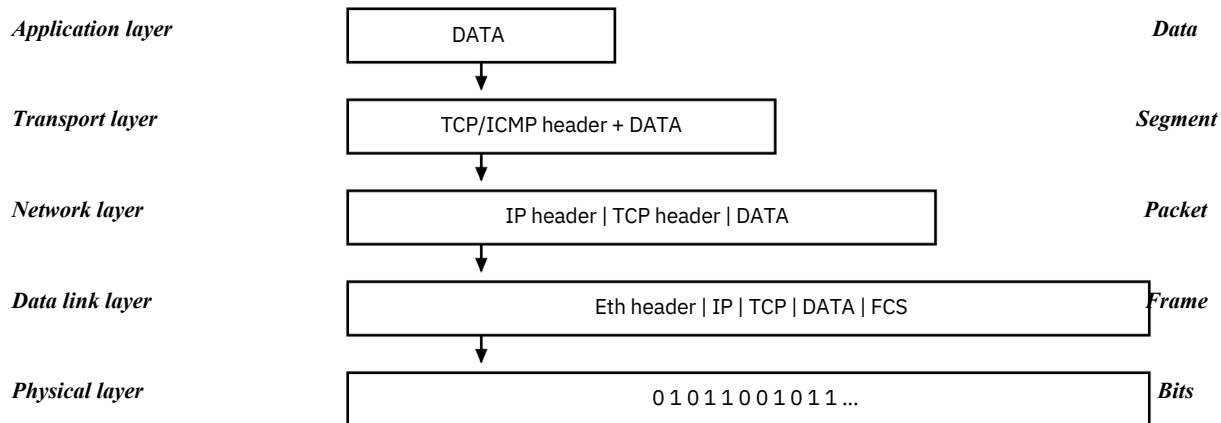
### **Features of Wireshark**

- Captures live packets from Ethernet, Wi-Fi, loopback and other interfaces and also reads saved .pcap files.
- Capture filters (before capture) and display filters (after capture).
- Colouring rules to identify the protocols easily (TCP, UDP, ICMP, ARP, HTTP ...).
- Shows all the protocol headers layer by layer with the raw bytes.
- Statistics tools like Conversations, Protocol Hierarchy and I/O Graph.

#### **Layer Header added**

| Layer       | Header added                                  | Protocol example | PDU name           |  |
|-------------|-----------------------------------------------|------------------|--------------------|--|
| Application |                                               | HTTP, DNS        | Data / Message     |  |
| Transport   | Port numbers, sequence no. (TCP)              | TCP, UDP         | Segment / Datagram |  |
| Network     | Source and destination IP address (20 B)      | IP, ICMP         | Datagram Packet    |  |
| Data Link   | Source and destination MAC (14 B) + FCS (4 B) | Ethernet         | Frame              |  |
| Physical    | None                                          | UTP, Wi-Fi       | Bits               |  |

### **DIAGRAM:**



**Fig 1: Encapsulation of data as it moves down the protocol stack**

## OUTPUT:

| Capturing from Ethernet - Wireshark |          |                   |                |          |      |                                                              |
|-------------------------------------|----------|-------------------|----------------|----------|------|--------------------------------------------------------------|
| No.                                 | Time     | Source            | Destination    | Protocol | Len  | Info                                                         |
| 1                                   | 0.000000 | 192.168.1.25      | 192.168.1.1    | DNS      | 70   | Standard query 0x3a1c A example.com                          |
| 2                                   | 0.012431 | 192.168.1.1       | 192.168.1.25   | DNS      | 86   | Standard query response 0x3a1c A example.com A 93.184.216.34 |
| 3                                   | 0.013102 | 192.168.1.25      | 93.184.216.34  | TCP      | 66   | 51322 > 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460              |
| 4                                   | 0.041875 | 93.184.216.34     | 192.168.1.25   | TCP      | 66   | 80 > 51322 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0            |
| 5                                   | 0.041932 | 192.168.1.25      | 93.184.216.34  | TCP      | 54   | 51322 > 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0                |
| 6                                   | 0.042210 | 192.168.1.25      | 93.184.216.34  | HTTP     | 142  | GET / HTTP/1.1                                               |
| 7                                   | 0.071655 | 93.184.216.34     | 192.168.1.25   | TCP      | 54   | 80 > 51322 [ACK] Seq=1 Ack=89 Win=65535 Len=0                |
| 8                                   | 0.093340 | 93.184.216.34     | 192.168.1.25   | HTTP     | 1294 | HTTP/1.1 200 OK (text/html)                                  |
| 9                                   | 1.402118 | 192.168.1.25      | 142.250.195.78 | ICMP     | 74   | Echo (ping) request id=0x0001, seq=21/5376, ttl=128          |
| 10                                  | 1.416875 | 142.250.195.78    | 192.168.1.25   | ICMP     | 74   |                                                              |
| 11                                  | 2.118004 | c4:e9:84:2a:5b:10 | Broadcast      | ARP      | 60   | Echo (ping) reply id=0x0001, seq=21/5376, ttl=117            |
|                                     |          |                   |                |          |      | Who has 192.168.1.25? Tell 192.168.1.1                       |

**Packet List pane - captured packets (colours show the protocol)**

| Packet Details - Frame 9 (ICMP Echo request)                                                                   |  |
|----------------------------------------------------------------------------------------------------------------|--|
| v Frame 9: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{8F3A21B0}, id 0 |  |
| Arrival Time: Oct 7, 2026 20:57:38.402118000 India Standard Time                                               |  |
| Frame Length: 74 bytes (592 bits)                                                                              |  |
| Protocols in frame: eth:ethertype:ip:icmp:data                                                                 |  |
| v Ethernet II, Src: 00:1a:2b:3c:4d:5e, Dst: c4:e9:84:2a:5b:10                                                  |  |
| Destination: c4:e9:84:2a:5b:10                                                                                 |  |
| Source: 00:1a:2b:3c:4d:5e                                                                                      |  |
| Type: IPv4 (0x0800)                                                                                            |  |
| v Internet Protocol Version 4, Src: 192.168.1.25, Dst: 142.250.195.78                                          |  |
| 0100 .... = Version: 4                                                                                         |  |
| .... 0101 = Header Length: 20 bytes (5)                                                                        |  |
| Total Length: 60                                                                                               |  |
| Identification: 0x7a31 (31281)                                                                                 |  |
| Time to Live: 128                                                                                              |  |
| Protocol: ICMP (1)                                                                                             |  |
| Source Address: 192.168.1.25                                                                                   |  |
| Destination Address: 142.250.195.78                                                                            |  |
| v Internet Control Message Protocol                                                                            |  |
| Type: 8 (Echo (ping) request)                                                                                  |  |
| Code: 0                                                                                                        |  |
| Checksum: 0x4d5a [correct]                                                                                     |  |
| Identifier (BE): 1 (0x0001)                                                                                    |  |
| Sequence Number (BE): 21 (0x0015)                                                                              |  |
| Data (32 bytes)                                                                                                |  |

| Packet Bytes - Frame 9 |                                                 |                   |
|------------------------|-------------------------------------------------|-------------------|
| 0000                   | c4 e9 84 2a 5b 10 00 1a 2b 3c4d5e08004500 00 3c | ...*[...+<M^...E. |
| 0010                   | 7a 31 00 00 80 01 00 00c0a801198efa c3 4e 08 00 | .<z1.....         |
| 0020                   | 4d 5a 00 01 00 15616263646566 67 68 69 6a 6b 6c | .N..MZ....abcdef  |
| 0030                   | 6d 6e 6f 70717273747576 77 61 62 63 64 65 66 67 | ghijklmnopqrstuv  |
| 0040                   | 68 69                                           | wabcdefghi        |

| Layer       | Bytes in the frame | Observed value                                                    |
|-------------|--------------------|-------------------------------------------------------------------|
| Ethernet II | 00 - 13 (14 bytes) | <del>c4:e9:84:2a:5b:10 &lt;- 00:1a:2b:3c:4d:5e, type 0x0800</del> |
| IPv4        | 14 - 33 (20 bytes) | 192.168.1.25 -> 142.250.195.78, TTL 128, protocol 1 (ICMP)        |
| ICMP        | 34 - 41 (8 bytes)  | Type 8, Code 0, id 1, seq 21                                      |
| Data        | 42 - 73 (32 bytes) | abcdefghijklmnopqrstuvwabcdefghi                                  |

**RESULT:**  
 Thus the features of Wireshark were studied. Packets were captured and the encapsulation of data at the Data link (Ethernet), Network (IP), Transport (TCP) and ICMP layers was observed in the packet details.